Verification packet: equations and statistics of the CBF-QP filter study

Al Mahdi Sallam (prepared for Dr. Brahmi)

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This packet lists the six items whose correctness the manuscript relies on. Each gives the statement, a derivation, the code that implements it and what to confirm. Item numbers match the sign-off table at the end. The final test was run on 25–26 October 2026 under a protocol the author locked without advisor review (commit `9baf6d7`); nothing below changes a reported number.

Setting

Unicycle robots, pose (x, y, θ), command $u = (v, \omega) \in \mathcal{U} = [-0.5, 0.5]\,\mathrm{m/s} \times [-1, 1]\,\mathrm{rad/s}$, held for $\Delta = 0.1\,\mathrm{s}$. Offset point $p = (x + \ell\cos\theta,\, y + \ell\sin\theta)$, $\ell = 0.05\,\mathrm{m}$, so $\dot p = J(\theta)u$ with $J = \begin{bmatrix}\cos\theta & -\ell\sin\theta \\ \sin\theta & \ell\cos\theta\end{bmatrix}$, $\det J = \ell$. Barriers $h_o = \|p - c\|^2 - D_o^2$, $D_o = R + \ell + R_o = 0.35\,\mathrm{m}$; $h_{ij} = \|p_i - p_j\|^2 - D_{rr}^2$, $D_{rr} = 2(R + \ell) = 0.30\,\mathrm{m}$. Since $\|x - p\| = \ell$, $h \geq 0$ implies the body discs (radius $R = 0.10\,\mathrm{m}$) do not overlap.

C1. Lemma 1 (one-sided sampled-data condition)

Statement. Let u be constant on $[t_k, t_k + \Delta]$, $\ddot h \geq -M$ there ($M \geq 0$), $\alpha\Delta \leq 1$, $h(t_k) \geq 0$. If $\dot h(t_k) + \alpha h(t_k) \geq \delta := M\Delta/2$, then $h(t_k + s) \geq (1 - \alpha s)h(t_k) \geq 0$ for all $s \in [0, \Delta]$.

Proof. By Taylor's theorem with the lower bound on $\ddot h$, $h(t_k + s) \geq h + \dot h\, s - \frac{M}{2}s^2$ (values at t_k). Using $\dot h \geq -\alpha h + M\Delta/2$,

$$h(t_k + s) \;\geq\; (1 - \alpha s)h + \tfrac{Ms}{2}(\Delta - s) \;\geq\; (1 - \alpha s)h \;\geq\; 0,$$

since $s \leq \Delta$ and $\alpha s \leq \alpha\Delta \leq 1$. □ Induction over periods gives $h \geq 0$ for all t provided every QP is feasible (assumption A9) and $h(0) \geq 0$ (A5). This is Theorem 3 of Breeden, Garg and Panagou, IEEE L-CSS 6:367–372 (2022), there with safe set $\{h \leq 0\}$ and $\gamma \in (0, 1]$, which corresponds to $\alpha\Delta \leq 1$. At $\alpha\Delta = 1$ the endpoint bound is $h(t_k + \Delta) \geq 0$, not strictly positive.
Code: `cbf_study/cbf_qp.py` (module docstring, `FilterConfig.validate` enforces $\alpha\Delta \leq 1$).

Please confirm: the Taylor step, the use of a one-sided bound, the induction under A5 and A9, and the mapping to Breeden et al. Thm. 3.

C2. The margins $\delta_o(d)$ and $\delta_{ij}(d)$ (eq. 4)

Under a held command, $\|\dot p\|^2 = v^2 + \ell^2\omega^2 \leq V_p^2$ with $V_p = \sqrt{0.5^2 + 0.05^2} \approx 0.5025\,\mathrm{m/s}$. Since $\dot\theta = \omega$ is constant, $\ddot p = \omega\, J'(\theta)u$ and $\|J'(\theta)u\| = \|\dot p\|$, so $\|\ddot p\| \leq \bar\omega V_p =: A_p$. Within the period $\|p(t) - c\| \leq d + V_p\Delta$, with $d = \|p(t_k) - c\|$.

$$\ddot h_o = 2\|\dot p\|^2 + 2(p - c)^\top \ddot p \quad\Rightarrow\quad |\ddot h_o| \leq 2V_p^2 + 2(d + V_p\Delta)A_p =: M_o(d).$$

For a pair, relative speed $\leq 2V_p$, relative acceleration $\leq 2A_p$, $\|p_i - p_j\| \leq d + 2V_p\Delta$: $|\ddot h_{ij}| \leq 8V_p^2 + 4(d + 2V_p\Delta)A_p =: M_{ij}(d)$. The implementation uses $\delta = M\Delta/2$ with these two-sided M; Lemma 1 needs only $\ddot h \geq -M$, for which $2(d + V_p\Delta)A_p$ (obstacles) suffices, so the $2V_p^2$ and $8V_p^2$ terms are extra conservatism (about 13× the tight one-sided obstacle margin at $d = D_o$; `results/design/margin_constants.json`). The row is $\dot h_o = g_o^\top u$ with $g_o = 2J^\top(p - c)$, i.e. $g_o^\top u \geq -\alpha h_o + \delta_o(d)$.

Code: `cbf_qp.margin_obstacle`, `cbf_qp.margin_robot`, `bmatch.barrier_rows`.

Please confirm: $\|\ddot{p}\| \le A_p$ under a held command; both bounds on $|\ddot{h}|$; that two-sided bounds are valid (if conservative) for the lemma.

C3. Equal split between robots

Robot i imposes $g_{ij}^\top u_i \ge -\frac{\alpha}{2} h_{ij} + \frac{1}{2}\delta_{ij}(d)$ with $g_{ij} = 2J_i^\top(p_i - p_j)$; robot j imposes the same with $i \leftrightarrow j$. Since $h_{ij} = h_{ji}$, d is symmetric, and $\dot{h}_{ij} = 2(p_i - p_j)^\top(\dot{p}_i - \dot{p}_j) = g_{ij}^\top u_i + g_{ji}^\top u_j$, adding the two rows gives $\dot{h}_{ij} \ge -\alpha h_{ij} + \delta_{ij}(d)$, the pair condition of Lemma 1. Required: both robots use the same state (A2–A3), apply commands over the same period (A4), the row is enabled for both (the range test is symmetric), and both QPs are feasible. In simulation the first two hold by construction (one call on the joint state), so "decentralised" refers to how the computation is split. Out-of-range pairs: a neighbour beyond 1.5 m needs at least 12 periods at $\le 2V_p\Delta \approx 0.10$ m per period to reach D_{rr}, and rows are rebuilt each period. In the final test 0.008% of robot-pair periods were beyond 1.5 m, so limited communication was not tested.

Please confirm: the summation argument and its conditions, and the out-of-range argument.

C4. Criterion (a) and the error-rate structure

Usefulness at density K requires (a), (b) and (c) (intersection–union test, so no split across components). Each component is tested one-sided at $\alpha_K = 0.05/3$, a Bonferroni split over the three density claims, giving familywise error ≤ 0.05 over the three usefulness claims. (a) is decided by the exact one-sided McNemar test: with n_{10} pairs failing off only and n_{01} failing on only, $p = P(X \ge n_{10})$, $X \sim \mathrm{Bin}(n_{10} + n_{01}, \frac{1}{2})$ (`locked_stats.mcnemar_one_sided_p`). (b) is decided by the Newcombe bound (C5) against the 5 pp margin; (c) by the bootstrap bound (C6) against 0.01 m.

Please confirm: the IUT logic, the Bonferroni denominator 3 (densities, not components), and the McNemar direction.

C5. Newcombe paired interval: method 8 versus method 10

With a, b, c, d the paired 2×2 counts, $p_1 = (a+b)/n$, $p_2 = (a+c)/n$, Wilson limits (l_i, u_i) and

$$\hat{\phi} = \frac{ad - bc}{\sqrt{(a+b)(c+d)(a+c)(b+d)}}, \qquad \theta - \sqrt{(p_1 - l_1)^2 - 2\hat{\phi}(p_1 - l_1)(u_2 - p_2) + (u_2 - p_2)^2}$$

is the lower limit (upper symmetric). This is Newcombe's method 8. Method 10 replaces $ad - bc$ by $\max(ad - bc - n/2, 0)$ when $ad - bc > 0$. The locked protocol labels its interval "method 10" but describes it as "without continuity correction", i.e. method 8; the code (function `newcombe_paired_ci` in `cbf_study/stats.py`, whose docstring also says "method 10") implements method 8. Recomputing every bound with method 10 changes it by < 0.001 pp and no criterion (`scripts/newcombe_method10_check.py`). The B-match comparison (Table 4) uses method 10. When a marginal variance is zero (B-match never fails), $\hat{\phi}$ is set to 0.

Please confirm: the method numbering against Newcombe (1998), the treatment of the inconsistency, and $\hat{\phi} = 0$ in the degenerate case.

C6. Criterion (c): formation estimand and bootstrap

For a pair with executed periods $T_{\mathrm{off}}, T_{\mathrm{on}}$, $L = \min(T_{\mathrm{off}}, T_{\mathrm{on}})$ and $\Delta e = \frac{1}{L}\sum_{t=1}^{L}[e_{\mathrm{on}}(t) - e_{\mathrm{off}}(t)]$, $e(t) = \frac{1}{3}\sum_r \|x_r(t) - \bar{x}(t) - s_r\|$; pairs with $L = 0$ are excluded (none occurred). Estimand: $\overline{\Delta e} = \frac{1}{9}\sum_p \mathrm{mean}_{i \in \mathcal{E}_p} \Delta e_i$. Bound: resample pairs with replacement within each actor, recompute the equal-actor mean, $B = 20{,}000$, seed 20261026 (`numpy.random.default_rng`), upper bound $= 1 - 0.05/3$ quantile (`method='linear'`); a normal bound from per-actor variances is a cross-check (function `formation_noninferiority` in `cbf_study/locked_stats.py`). Intervals reflect layout sampling for nine fixed actors from three training runs, not variation across training runs.

Please confirm: the estimand, the stratified resampling and the interpretation of the interval.

Additional: infeasibility (exploratory, for information)

At all 433 infeasible robot states, the binding row sat at its stand-off $h^\star = \delta/\alpha$ ($|h - h^\star| \le 4.3 \times 10^{-8}\,\mathrm{m}^2$). An exact vertex enumeration of $t^\star = \max_{u \in \mathcal{U}} \min_r (n_r^\top u - c_r)$ gives $t^\star \in [-6.7 \times 10^{-7}, -5.6 \times 10^{-10}]$; every minimal conflicting set has three rows (407: two neighbour + one obstacle; 26: one neighbour + two obstacles), never the command box; 7 cases lie within the solver tolerance 10^{-9}. (`scripts/infeasibility_conflicts.py`.) A comment on whether this "three-sided pinch at the stand-off" reading is sound would help.

Sign-off

Item	Confirmed (Y/N)	Comments
C1 Lemma 1 and induction		
C2 Margins δ_o, δ_{ij}		
C3 Equal split and ranges		
C4 IUT, Bonferroni, McNemar		
C5 Newcombe method 8 / 10		
C6 Formation estimand, bootstrap		
Margins 5 pp and 0.01 m acceptable?		

Reviewer: _____ Date: _____